THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture – Forward & futures pricing

Dr Xiaoting Wei

Outline

- Introduction
  - **Consumption vs Investment assets**
  - Measuring interest rates
- Investment assets
  - No income
  - Known income (bond futures)
  - Known yield (stock index futures)
- Consumption assets

- **Reading:** Chapter 5

Consumption Vs. Investment Assets

Investment assets are assets held by significant numbers of people purely for investment purposes.

- Examples of investment assets are stocks and bonds.

Consumption assets are assets held primarily for consumption and not usually for investment purposes.

- Examples of consumption assets are commodities such as copper and oil.
- Gold and silver are both investment and consumption assets. For futures and forward pricing, they are treated as investment assets.

We can **use arbitrage arguments** to determine the forward and futures price of an investment asset from its spot price and other observable market variables. We **cannot do this for consumption assets**.

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Measuring Interest Rates

In this subject, we will mainly use continuous interest rate.

- As such you will need to ensure that your interest rates are expressed as continuously compounded interest rates.
- We will detail any exceptions to this rule as we progress through the subject.

For Example:

- \$100 grows to $\$100e^{rt}$ when invested at a continuously compounded rate r for time t.
- \$100 received at time t discounts to $\$100e^{-rt}$ at time zero when the continuously compounded discount rate is r.

Introduction: Measuring Interest Rates

	Future Value	Present Value
$FNCE10002$ Principles of Finance	$FV = PV(1+r)^t$	$PV = \frac{FV}{(1+r)^t}$
$FNCE30007$ Derivative Securities	$FV = PVe^{rt}$	$PV = FVe^{-rt}$

Converting annual nominal rate to continuously compounded rate:

$$r_c = m \cdot ln\left(1 + \frac{r_m}{m}\right)$$

where m is number of times the interest rate is compounded in a year, $ln(\cdot)$ is the natural logarithm function, r_c is a rate of interest with continuous compounding, and r_m is the nominal rate compounded m times per annum.

Introduction: Measuring Interest Rates

Example:

If a nominal rate of 10% p.a. is compounded quarterly, what is the equivalent continuous rate?

$$r_c = 4 \cdot ln\left(1 + \frac{10\%}{4}\right)$$

$$r_c = 9.88\%$$

If a nominal rate of 10% p.a. is compounded continuously what is the effective rate?

$$e^r - 1$$

$$e^{0.10} - 1$$

$$2.71828\ldots^{0.10} - 1 = 10.51\%$$

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Assumptions

In this lecture we make the following assumptions regarding market participants:

1. They are subject to no transaction costs when they trade;
2. They are subject to the same tax rate on all net trading profits;
3. They can borrow money at the same risk-free rate of interest as they can lend money; and,
4. They take advantage of arbitrage opportunities as they occur.

Forward and Futures Contract Prices – Cost of Carry Model

- It is well known in practice that if interest rates are constant, a futures contract has the same value as an otherwise identical forward contract.

- That is, although a futures contract has a complicated cash flow pattern (via the marking to market feature) it can be valued as though it were a forward contract.

- Since a forward contract has only a single cash flow, it is easy to value.

- Consequently, it is industry practice to value futures contracts as though they were forward contracts.

Forward and Futures Contract Prices – Cost of Carry Model

Consider the strategy of:

- Borrowing enough money to buy one unit of an investment asset that provides the holder with no income, and has no holding costs. Non-dividend paying stocks and zero-coupon bonds are examples of such investment assets. The borrower incurs the obligation to pay for the associated interest through time *T;* and,
- Entering into a forward or futures contract to sell the asset at time *T*.

The value of this position in terms of the initial (time 0) and terminal (time *T*) cash flows is tabulated in the following table.

Forward and Futures Contract Prices – Cost of Carry Model

Arbitrage Relationship Between Spot and Forward Contracts

Position	**Initial Cash Flow**	**Terminal Cash Flow**
Borrow	S_0	$-S_0e^{rT}$
Buy one unit of asset	$-S_0$	S_T
Enter 6-month forward sale	0	$F-S_T$
Net portfolio value	***0***	$F- S_0e^{rT}$

$$F - S_0e^{rt} = 0$$
$$F = S_0e^{rt}$$

Forward and Futures Contract Prices – Cost of Carry Model

Example:

- Consider a four-month forward contract to buy a zero-coupon bond that will mature one year from today.

- The current price of the bond is \$930. We assume that the four-month risk-free rate of interest (continuously compounded) is 6% per annum. The forward price, F_0, is given by:

$$F_0 = 930e^{0.06\times\frac{4}{12}} = \$948.79$$

Arbitrage

In general if:

$$F_0 > S_0 e^{rT}$$

- Arbitrageurs can make a riskless profit from buying the asset and entering into a short forward contract on the asset. This strategy is financed by borrowing funds at the risk free-rate of interest.

$$F_0 < S_0 e^{rT}$$

- Arbitrageurs can make a riskless profit by shorting the asset and entering into a long forward contract. The excess funds are invested at the risk-free rate of interest until they are needed to buy back the asset.

Forward and Futures Contract Prices on Assets with Known Income

We now consider a forward contract on an investment asset that will provide a perfectly predictable cash income to the holder.

- Examples: Stocks paying known dividend amount and coupon-bearing bonds.

$$F_0 = (S_0 - I)e^{rT}$$

- Where I is the **present value** of the income.

Forward and Futures Contract Prices on Assets with Known Income

$$F_0 = (S_0 - I)e^{rT}$$

F_0 is now lower than it would be if the underlying asset generated no income because:

- The holder of the underlying asset receives the income *I* and is therefore willing to sell the asset for less.
- The buyer who will acquire the asset in the future will miss out on the income *I* and will therefore be willing to pay less for it.

Forward and Futures Contract Prices on Assets with Known Income

Example:

- Consider a long forward contract to purchase a coupon-bearing bond whose current price is $900. We will assume that the forward contract matures in nine months. We also assume that a coupon payment of $40 is expected after 4 months. The four-month and nine-month risk-free interest rates continuously compounded are 3% and 4% per annum, respectively.

Forward and Futures Contract Prices on Assets with Known Income

$$I = 40e^{-0.03\times\frac{4}{12}} = 39.60$$

$$F_0 = (900 - 39.60)e^{0.04\times0.75} = \$886.60$$

Forward and Futures Contract Prices on Assets with Known Yield

How do we deal with a situation where the asset underlying a forward contract provides a known yield rather than known cash income?

- A yield implies that the known income is expressed as a percentage of the asset's price at the time the income is paid.
- We define q as the average yield per annum on an asset during the life of a forward contract with continuous compounding.
- The formula we use is:

$$F_0 = S_0 e^{(r-q)T}$$

Forward and Futures Contract Prices on Assets with Known Yield

Example: Consider a six-month forward contract on an asset that is expected to provide an income equal to 2% of the asset price once during a six-month period. The risk-free rate of interest with continuous compounding is 10% per annum. The asset price is $25.

The yield is 4% per annum with semi-annual compounding. Converting this to continuous compounding we get:

$$r_c = m \ln\left(1 + \frac{r_m}{m}\right) = 2 \ln\left(1 + \frac{0.04}{2}\right) = 0.0396$$

So the forward price is given by:

$$F_0 = 25e^{(0.10-0.0396)\times 0.5} = \$25.77$$

Forward and Futures Contract Prices on Stock Indices

A stock index can be viewed as an investment asset paying a dividend yield.

The futures price and spot price relationship is therefore:

$$F_0 = S_0 e^{(r-q)T}$$

- Where q is the dividend yield on the portfolio represented by the index.
- Remember, with indices, they are stated as points. Therefore, the number of points must be multiplied by a factor to end up with a dollar value for the contract. In Australia, the convention is $25 per point.

Forward and Futures Contract Prices on Stock Indices

Consider a 1-year futures contract on the ASX S&P200. Suppose that the stocks underlying the index provide a dividend yield of 5% per annum continuously compounded, that the current value of the index is 3529, and that the continuously compounded risk-free interest rate is 10% per annum.

$$F_0 = S_0 e^{(r-q)T}$$

$$F_0 = 3529 e^{(0.10-0.05)} = 3710$$

If we multiply this value by \$25 per point, each futures contract will hedge a dollar value of \$92,750.

Forward and Futures Contract Prices on Stock Indices

In general if:

$$F_0 > S_0 e^{(r-q)T}$$

- An arbitrageur can make a riskless profit by buying the stocks underlying the index and shorting index futures contracts. This strategy will be financed by borrowing funds at the risk-free interest rate.

$$F_0 < S_0 e^{(r-q)T}$$

- An arbitrageur can make a riskless profit by shorting the stocks underlying the index and taking a long position in index futures contracts. The excess funds will be invested at the risk-free interest rate until needed to buy back the stocks.

Futures and Forward on Currencies

The underlying asset in a forward or futures currency contract is a certain number of units of a foreign currency.

A foreign currency is analogous to a security providing a dividend yield. A foreign currency has the property that the holder of the currency can earn interest at the risk-free interest rate prevailing in the foreign country. For example, the holder can invest the foreign currency in a foreign-denominated bond.

Thus, the continuous dividend yield is the foreign risk-free interest rate.

It follows that if r_f is the foreign risk-free interest rate, S_0 as the current spot price in dollars of one unit of the foreign currency and F_0 as the forward or futures price in dollars of one unit of the foreign currency.

$$F_0 = S_0 e^{(r - r_f)T}$$

Futures and Forward on Currencies

Example:

The spot exchange rate is 1EUR=1.12 USD. The 6-month USD interest rate is 6% p.a. continuously compounded, and the EUR rate is 3% p.a. continuously compounded. What is the 6-month forward rate? How much USD is needed to buy EUR 1,000,000 under this forward?

USD is domestic currency, and EUR is foreign currency, so

$$F_0 = S_0 e^{(r-r_f)T}$$

$$F_0 = 1.12e^{(0.06-0.03)0.5} = 1.1369$$

USD cost = 1,000,000 × 1.1369 = $1,136,900

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Forwards and Futures on Commodities

Cost of carry (u) of the underlying commodity.

- This is the cost of holding a physical quantity of the commodity. For wheat the cost of carry is the storage cost; for live hogs it consists of storage and feed costs.
- Holding a stock index provides the benefit of receiving dividends. The dividends are a negative cost of carry.

Forwards and Futures on Commodities

First, let us consider the futures prices of commodities that are investment assets such as gold and silver.

In the absence of storage costs and income the forward price of a commodity that is an investment asset is given by:

$$F_0 = S_0 e^{rT}$$

If there are storage costs, U is the present value of all of the storage costs less all income during the life of the forward contract, and the forward price is given by:

$$F_0 = (S_0 + U)e^{rT}$$

Forwards and Futures on Commodities

If storage costs and income are given as a percentage, then u is the percentage storage costs less the percentage income during the life of the forward contract, and the forward price is given by:

$$F_0 = S_0 e^{(r+u)T}$$

Forward and Futures on Commodities

Now let us consider Consumption Commodities:

- Commodities that are consumption assets rather than investment assets usually provide no income, but can be subject to significant storage costs.
- Individuals and companies who keep such a commodity in inventory do so because of its consumption value, not because of its value as an investment.
- They are reluctant to sell the commodity and buy forward contracts because forward contracts cannot be consumed.
- There is therefore nothing to keep the previous equations holding (i.e. arbitrage).

Forward and Futures on Commodities

As a result:

- Due to the high storage costs of consumption commodities, U is the present value of all of the storage costs, and the forward price is given by:

$$F_0 \leq (S_0 + U)e^{rT}$$

- If storage costs are expressed as a proportion u of the spot price, the equivalent formula is:

$$F_0 \leq S_0 e^{(r+u)T}$$

Forward and Futures on Commodities

The reason we do not have equality in the formula's on the previous slide is because users of a consumption commodity may feel that ownership of the physical commodity provides benefits that are not obtained by holders of futures contracts.

- For example, an oil refiner is unlikely to regard a futures contract on crude oil in the same way as crude oil held in inventory.
- The crude oil in inventory can be used in the refining process whereas a futures contract cannot.

Forward and Futures on Commodities

The benefits from holding the physical asset is referred to as the convenience yield.

As such we can re-write the equations on the previous slide, where y, the convenience yield simply measures the extent to which the left hand side is less than the right hand side in those previous equations:

$$F_0 e^{yT} = (S_0 + U)e^{rT}$$

And if the storage costs are a percentage, then:

$$F_0 e^{yT} = S_0 e^{(r+u)T} or\ F_0 = S_0 e^{(r+u-y)T}$$

Valuing Forward Contracts

The value of a forward contract at the time it is first entered into is zero. At a later stage it may prove to have a positive or negative value.

Suppose that

- K is delivery price in a forward contract (the initial forward price when the contract was negotiated some time ago);
- F is the current forward price for the contract that was negotiated some time ago;
- The delivery date is T years from today;
- r is the T-year risk-free interest rate; and,
- f is the value of the forward contract today.

The value of a long forward contract (on both investment and consumption assets, ƒ, is:

$$f = (F - K)e^{-rT}$$

Similarly, the value of a short forward contract is

$$f = (K - F)e^{-rT}$$

Valuing Forward Contracts

Example:

- A long forward contract on a non-dividend paying stock was entered into some time ago.
- It currently has six months to maturity.
- The risk-free rate of interest (with continuous compounding) is 10% per annum.
- The stock price is \$25, and the delivery price is \$24.
- What is the value of the forward contract?

- $F_0 = 25e^{0.1\times0.5} = \$26.28$

$$f = (26.28 - 24)e^{-0.1\times0.5} = \$2.17$$

Valuing Futures Contracts

The value of a forward contract at the time it is first entered into is also zero.

Futures are marked to market daily, so its value can be realised on the day, not until T.

Value for long futures

$$f = F - K$$

Value for short futures

$$f = K - F$$

Next Week

- **Introduction to Options**
  - Option types and positions
  - Moneyness, intrinsic and time value
  - Factors affecting option prices
  - Upper and lower bounds
  - Put-call parity
  - Early exercise
  - The effect of dividends